

UT Austin Master's Thesis: Holographic Tensor Networks: Computational Insights into Quantum Gravity and Quantum Information

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1 Master's Thesis: Holographic Tensor Networks

1.1 General Outline

This research program explores the computational connections between tensor networks, quantum gravity, quantum information, and quantum many-body systems through a systematic three-phase approach.

Phase 1: Build Foundation & Reproduce (*3–4 months*)

Goal: Establish computational infrastructure and reproduce known results

- **Infrastructure Development**

- Configure research repository and build analysis tools
- Establish modular codebase architecture
- Integrate cloud computing backends (Azure/AWS)

- **Tensor Network Implementations**

- Implement PyTorch + TensorNetwork framework for:
 - * Matrix Product States (MPS)
 - * Projected Entangled Pair States (PEPS)
 - * Multi-scale Entanglement Renormalization Ansätze (MERA)
 - * Hyper-invariant Tensor Networks
 - * Hybrid Neural Tensor Networks
- Apply to various quantum systems (Ising model in various dimensions, critical systems)
- Develop efficient data generation modules for ML-based tensor network state representation
 - * Consider leveraging VQEs for data generation of complex systems

- **Quantum Algorithms & Quantum Machine Learning**

- Implement canonical algorithms: Grover's search, VQEs, QCNNs

- Develop on simulators (`cirq.Simulator`)
- Deploy on physical QPUs (IonQ, Quantinuum) via AWS Braket/Azure Quantum
- Leverage libraries like `Cirq`, `PennyLane`, etc.

- **Milestone: End of January 2026**

- Complete research repository structure:
 - * `/tensor_networks`: Core implementations
 - * `/quantum_algorithms`: Canonical quantum algorithms
 - * `/quantum_machine_learning`: QML algorithms
 - * `/quantum_backends`: Cloud integration
 - * `/utils`: Reusable modules
 - * `/references`: Relevant research
- Seamless integration with serverless training (Google Colab)
- Reproduced results: entanglement scaling, Ising criticality, Ryu-Takayanagi formula
- Consider publishing models and datasets to Hugging Face for ease of use

Phase 2: Systematic Study & Peripheries (*3–4 months*)

Goal: Generate novel insights through systematic computational studies

- **Bottom-Up Approaches**

- Train MERA on multiple quantum systems
- Systematically extract emergent geometric properties
- Compare architectural variations across different systems
- *Target publication*: “Computational Study of Emergent Holography”
 - * Inspired by scaling law studies (e.g., Kaplan et al. for Transformers)

- **Top-Down Approaches**

- Enforce geometric constraints on network architectures
- Analyze boundary entanglement properties of constrained networks

- **Connections to Loop Quantum Gravity**

- Explore relationships between LQG spin networks and tensor networks
- Investigate LQG-holography connections through computational lens

- **Milestone: April/May 2026**

- Novel computational insights documented in `/research`
- LaTeX documentation for findings
- Draft publication materials

Phase 3: Novel Directions (*4–6 months*)

Goal: Pursue original research directions for thesis completion

- **Academic Timeline**

- Summer 2026: Thesis Credit A
- Fall 2026: Thesis Credit B
- Target graduation: December 2026

- **Potential Research Directions**

- Emergent geometries from dynamic MERA architectures
- ML-guided architecture search for holographic error-correcting codes
- Tensor network representations of spin networks and spin foams
- Transfer learning between quantum systems
- Holographic-inspired neural network architectures
- Quantum optimization scaling laws
- Progeny of any other foundational efforts

- **Deliverable**

- Master’s thesis presenting novel research at the intersection of tensor networks, quantum information, and quantum gravity
- Target: publishable results in computational quantum physics



Figure 1: Rango